

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5112 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I Logeshwari S(192565040) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

SIGNATURE OF THE STUDENT :
Logeshwari S

QUESTION 1: SSL/TLS Overhead and Web Server Performance Optimization

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Answer

Given:

- Payload per response = 2000 bytes
- TLS record overhead = 25 bytes
- Requests per second = 10,000
- Handshake overhead = 3 KB per session
- New TLS handshakes = 1,000 per second

A. Calculate the total data transmitted per second including TLS overhead.

Data per request = $2000 + 25 = 2025$ bytes.

For 10,000 requests: $2025 \times 10,000 = 20,250,000$ bytes/s.

Handshake data = $1,000 \times 3 \text{ KB} = 3,000 \text{ KB/s} = 3,000,000 \text{ bytes/s}$ (using $1 \text{ KB} = 1000$ bytes).

Total data = $20,250,000 + 3,000,000 = 23,250,000 \text{ bytes/s} \approx 23.25 \text{ MB/s} \approx 186 \text{ Mbps}$.

B. Determine the percentage overhead due to TLS encryption.

Record overhead = $25 \times 10,000 = 250,000$ bytes/s.

Percentage record overhead = $(250,000 / 20,000,000) \times 100 = 1.25\%$.

If session-handshake overhead is also included, total TLS overhead = 3,250,000 bytes/s and percentage overhead relative to payload = $(3,250,000 / 20,000,000) \times 100 = 16.25\%$.

C. Estimate bandwidth consumed due to handshake overhead per second.

Handshake bandwidth = 3,000,000 bytes/s $\times 8 = 24,000,000$ bits/s = 24 Mbps.

D. Analyze the trade-off between security and latency, and propose optimization techniques.

TLS 1.3 provides strong confidentiality, integrity, authentication, and protection against eavesdropping and tampering. However, encryption adds record overhead and new handshakes consume CPU, bandwidth, and latency. A high rate of new handshakes can therefore reduce server performance.

- Use TLS session resumption/PSK to avoid repeating full handshakes.
- Use HTTP/2 to multiplex multiple requests over fewer connections.
- Use HTTP/3 with QUIC to reduce connection-establishment latency and improve performance on changing networks.
- Use connection pooling and keep-alive to reduce repeated handshakes.
- Monitor CPU, bandwidth, handshake rate, and latency to tune the server.

Conclusion: TLS overhead is relatively small for individual records, but frequent handshakes can become a significant bandwidth and latency cost. Reusing secure sessions provides a good balance between security and performance.

QUESTION 2: PGP-Based Email Encryption Load in Enterprise System

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Answer

Given:

- Attachment size = 10 MB per email
- PGP overhead = 800 bytes per email
- Emails per day = 5,000

A. Calculate the total data transmitted per day including overhead.

Assuming 10 MB = 10,000,000 bytes:

Data per email = 10,000,000 + 800 = 10,000,800 bytes.

Daily data = 10,000,800 × 5,000 = 50,004,000,000 bytes ≈ 50.004 GB/day.

Thus, the PGP overhead adds 4,000,000 bytes, or approximately 4 MB per day.

B. Determine the percentage increase in storage due to PGP.

Percentage increase = $(800 / 10,000,000) \times 100 = 0.008\%$.

Therefore, PGP causes only a very small storage increase for a 10 MB attachment.

C. Analyze the computational overhead during encryption/decryption.

PGP normally uses a hybrid approach: a symmetric cipher encrypts the bulk data, while public-key cryptography protects the session key and digital signatures provide authentication and integrity. Symmetric encryption of large attachments is efficient, whereas public-key operations are more computationally expensive. Hashing and signature verification also consume CPU resources. At 5,000 emails per day, the server

must handle both encryption and decryption workloads and key-management operations.

D. Recommend optimization strategies.

- Compress data before encryption when compression is beneficial.
- Use efficient symmetric encryption for large attachments and reserve public-key operations for key protection/signatures.
- Batch or queue non-urgent encryption jobs to smooth CPU usage.
- Use hardware cryptographic acceleration where supported.
- Use caching and efficient key management to reduce repeated public-key operations.

Conclusion:

PGP adds negligible storage overhead in this scenario, but computational cost can become important as email volume grows.

QUESTION 3: IPSec Transport vs Tunnel Mode Performance Comparison

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header. Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Answer

Given:

- Payload = 1000 bytes
- Original header = 20 bytes
- Original packet = 1020 bytes
- Transport-mode overhead = 30 bytes
- Tunnel-mode overhead = 50 bytes
- Traffic = 20,000 packets/s

A. Compute total packet size for both modes.

Transport mode = $1020 + 30 = 1050$ bytes.

Tunnel mode = $1020 + 50 = 1070$ bytes.

B. Calculate bandwidth usage for each mode per second.

Transport mode = $1050 \times 20,000 = 21,000,000$ bytes/s = 21 MB/s = 168 Mbps.

Tunnel mode = $1070 \times 20,000 = 21,400,000$ bytes/s = 21.4 MB/s = 171.2 Mbps.

C. Compare efficiency and security benefits of both modes.

Transport mode has lower overhead and is therefore more bandwidth-efficient. It protects the payload of the original IP packet while retaining the original IP header. It is commonly suitable when communicating directly between security-aware hosts.

Tunnel mode has higher overhead because it encapsulates the original IP packet inside a new IP packet. It provides stronger network-level isolation and is especially useful for VPNs, gateways, and site-to-site communication.

D. Recommend the appropriate mode for internal vs external communication.

For controlled internal host-to-host communication, transport mode is generally appropriate because it provides security with lower overhead. For external communication such as site-to-site VPNs or gateway-to-gateway traffic, tunnel mode is generally preferred because it protects the complete original IP packet and hides internal addressing information.

Conclusion:

Transport mode provides better efficiency, while tunnel mode provides stronger encapsulation and is more suitable for VPN-style external communication.

QUESTION 4: S/MIME Certificate Management in Large Organization

An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Answer

Given:

- Employees/certificates = 12,000
- Certificate size = 2 KB
- Old and new certificates coexist during renewal = 2 certificates per employee

A. Calculate total storage required during the renewal period.

Storage for one set = $12,000 \times 2 \text{ KB} = 24,000 \text{ KB} = 24 \text{ MB}$.

During renewal, two sets are stored: $24 \text{ MB} \times 2 = 48 \text{ MB}$.

Therefore, total certificate storage during the one-month renewal period is approximately 48 MB.

B. Estimate certificate distribution overhead across the network.

If one 2 KB certificate is distributed to each of 12,000 employees, one complete certificate distribution cycle requires $24,000 \text{ KB} = 24 \text{ MB}$ of certificate data. If both old and new certificates are distributed during the renewal process, the total is approximately 48 MB. Actual network usage may be higher because of protocol headers, authentication, retries, and directory/PKI traffic.

C. Analyze challenges in certificate lifecycle management at scale.

- Issuing, renewing, revoking, and replacing certificates for 12,000 users is operationally complex.
- Expired certificates can interrupt secure email communication.
- Lost or compromised private keys require rapid revocation and replacement.
- Different users and devices may have different certificate stores and software.
- The organization must maintain accurate identity-to-certificate mapping and audit records.

D. Propose optimization strategies.

- Automate certificate enrollment, renewal, and revocation.
- Use a centralized PKI with clear certificate policies and role-based administration.
- Use monitoring and alerts before certificates expire.
- Maintain centralized certificate inventory and audit logs.
- Use short-lived certificates when appropriate, combined with reliable automated renewal.

Conclusion:

The storage requirement is modest, but lifecycle management is the major challenge. Automation and centralized PKI significantly improve reliability and scalability.

QUESTION 5: Email Encryption Latency in Cloud-Based Systems

A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Answer

Given:

- AES time = 0.2 ms/MB
- Attachment size = 8 MB
- RSA key-exchange time = 5 ms/email
- Emails = 8,000 per hour

A. Calculate total encryption time per email.

AES time = $0.2 \times 8 = 1.6$ ms.

RSA time = 5 ms.

Total encryption time per email = $1.6 + 5 = 6.6$ ms.

B. Estimate total processing time per hour.

Total time = $6.6 \text{ ms} \times 8,000 = 52,800 \text{ ms} = 52.8$ seconds per hour.

Thus, the theoretical cryptographic processing time is 52.8 seconds per hour, excluding I/O, network delays, queuing, software overhead, and other system operations.

C. Analyze the performance bottleneck between symmetric and asymmetric operations.

AES requires 1.6 ms per email, while RSA requires 5 ms. RSA therefore consumes $5 / 6.6 \times 100 \approx 75.76\%$ of the cryptographic time. AES consumes approximately 24.24%. Hence, RSA key exchange is the dominant cryptographic bottleneck in this scenario.

D. Recommend optimization strategies.

- Use efficient modern key-establishment mechanisms such as ECC-based cryptography where appropriate.
- Use parallel processing so multiple email encryption jobs can run concurrently.
- Use hardware cryptographic acceleration when available.
- Reuse secure sessions/keys appropriately rather than performing expensive public-key operations unnecessarily.
- Monitor CPU utilization, throughput, queue length, and latency to identify real bottlenecks.

Conclusion:

The total cryptographic time is 6.6 ms per email and 52.8 seconds for 8,000 emails. RSA is the major contributor to encryption latency, so optimizing asymmetric operations can provide the greatest performance improvement.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation